

HeidiSQL Quick Guide

A practical beginner reference for working with MariaDB through HeidiSQL

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Key distinction

MariaDB is the database engine. HeidiSQL is the client interface. HeidiSQL sends SQL to the database and displays the results, warnings, and errors.

1. What HeidiSQL is useful for

- Connecting to MariaDB/MySQL and other supported database systems
- Browsing databases, tables, columns, keys, and indexes
- Writing and executing SQL
- Viewing result grids and warnings
- Importing and exporting data
- Creating and altering tables
- Inspecting relationships and data quickly

2. A safe working rhythm

1. Confirm you are connected to the correct session and database.
2. Read the SQL before you run it.
3. Run SELECT checks before UPDATE or DELETE statements.
4. After imports, read Records, Skipped, Warnings, and Affected rows.
5. Validate the data, not just the absence of errors.

3. Executing SQL

Use the query editor for SQL statements. Keep explanatory notes outside executable SQL, or prefix comments with `--` or wrap them in `/* ... */`.

```
-- This is a valid SQL comment.
```

```
SELECT COUNT(*)  
FROM employees;
```

```
/*  
This is also a valid  
multi-line SQL comment.  
*/
```

4. Importing CSV data

HeidiSQL can import through its interface, or you can use MariaDB's LOAD DATA LOCAL INFILE for more controlled transformations.

Import issue	What to check
Records = 0	Wrong line ending, wrong number of header rows, wrong file, or parser setting.
File not found	Exact local path and filename. Do not trust a renamed upload filename.
ANSI/encoding locked	Check whether the import dialog switched to client-side parsing.
Many truncation warnings	Source values do not fit the target type or need explicit conversion.
Blank dates become 0000-00-00	Load through a variable and convert blank to NULL.

5. Server parses vs client parses

In HeidiSQL's text import workflow, the parser choice can change available encoding and import behavior.

- Server parses file contents: uses database-side loading behavior such as LOAD DATA and usually gives more control for large structured files.
- Client parses file contents: HeidiSQL reads/parses the file and may expose different encoding or transformation behavior.
- If the import dialog suddenly looks different or encoding options disappear, check which parser mode is selected.

6. Line endings

`LINES TERMINATED BY '\r\n'` -- common Windows CRLF

`LINES TERMINATED BY '\n'` -- common LF

A wrong line-ending choice can produce Records: 0 even with Skipped: 0 and Warnings: 0.

7. Header rows

`IGNORE 1 LINES` -- one header row

`IGNORE 2 LINES` -- title row + header row

Count what is actually in the source file. Do not assume all CSVs from the same project use the same number of header rows.

8. Reading the execution log

Message	Interpretation
Records	How many source records were processed by the load.
Skipped	Rows not inserted.
Warnings	Potential conversion, truncation, or data issues.
Affected rows	Rows changed by the statement.
Found rows	Rows returned/found depending on statement type.

9. Common errors

- SQL Error 1064: syntax error. Look near the location MariaDB names, but also inspect the line immediately before it.
- Data truncated: inspect the raw source value and the target data type.
- Cannot add foreign key: find unmatched values or incompatible column definitions.
- Duplicate entry: determine whether you have a true duplicate, an incorrect key, or a repeated import.
- File not found: the path MariaDB/HeidiSQL is using does not match the local file.

10. Useful habits

- Keep raw source files unchanged.
- Use staging tables for intentionally messy sources.
- Move successfully loaded source files to a Loaded folder if that helps you track progress.
- Save important transformation and validation SQL as scripts.
- Run a validation pass after a batch of imports.
- Do not treat Warnings: 0 as success if Records: 0.

11. HeidiSQL vs MariaDB

If the issue is...	Usually look at...
SQL syntax	MariaDB
Table rules / foreign keys	MariaDB
Import dialog layout	HeidiSQL
Encoding options in the dialog	HeidiSQL parser/import mode
Query result values	MariaDB + your SQL
Old warnings still visible in the log	HeidiSQL interface/log history